

CeTZ Gallery

Diagrams in idiomatic CeTZ 0.5.2, each chosen for a technique worth copying.

[View as PDF](#). [View as HTML](#). [View source](#).

Every figure is a CeTZ redrawing of a TikZ figure from Walmes M. Zeviani's *Tikz Gallery* [1]. Each entry links its original. The redrawings translate the labels from Portuguese to English, use Typst's built-in colors, and differ in detail.

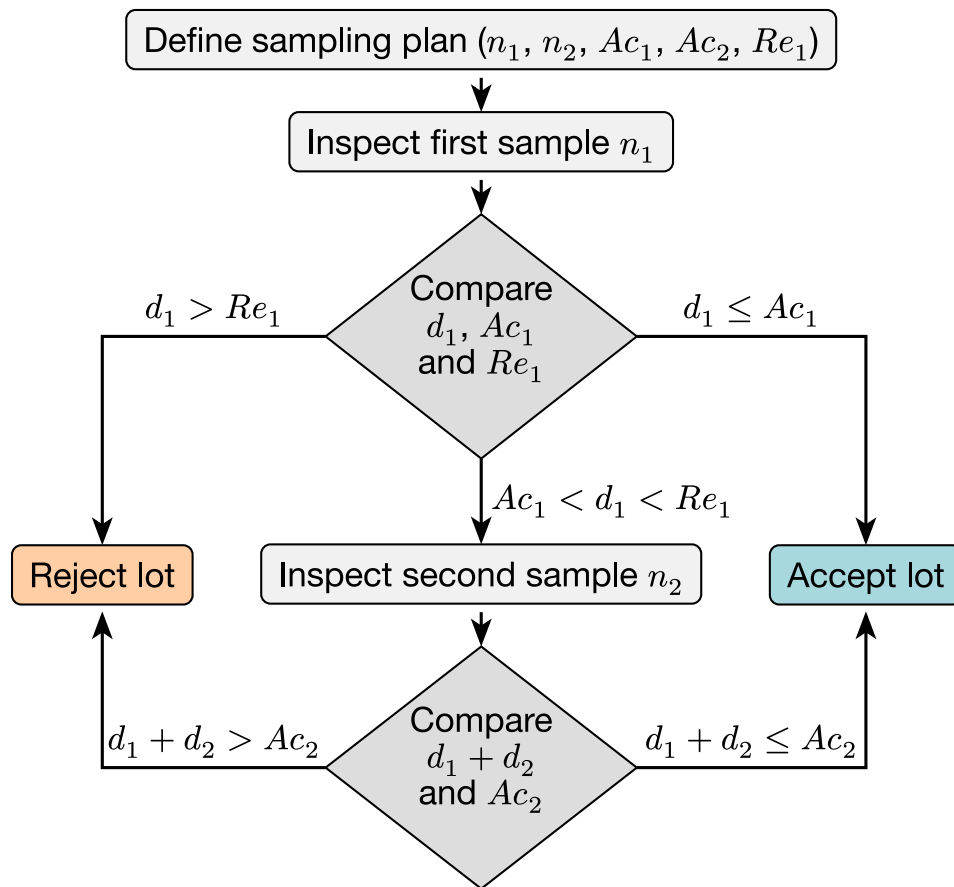
[1] Walmes M. Zeviani, *Tikz Gallery*. <https://github.com/walmes/Tikz>, <http://leg.ufpr.br/~walmes/Tikz/>.

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Sampling-plan flowchart

Redrawn from [sampling-plan-2.pgfc \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*. Relative placement from anchors; diamond decisions built from a text box's anchors; elbow branches with labels; two arrows meeting one box.



```

#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

// Double-sampling acceptance plan: Ac = acceptance number, Re = rejection number.
#let Ac = $A c$
#let Re = $R e$

#cezt.canvas({
  import cetz.draw: *
  set-style(stroke: 0.6pt + black)

  // A Typst box gives rounded corners while the anchors stay on its bounding box.
  let step(pos, name, body, fill: silver.lighten(60%), anchor: "north") = content(pos, name:
name,
  anchor: anchor, box(body, fill: fill, stroke: 0.6pt + black, radius: 3pt, inset: 5pt))
  // A diamond whose edge midpoints touch the corners of the text box.
  let decision(pos, name, body) = group(name: name, anchor: "north", {
    content(pos, box(width: 4.5em, align(center, par(leading: 0.35em, body))), name: "text")
    on-layer(-1, line(..("north", "east", "south", "west").map(a => ("text", 200%, "text." +
a)),
      close: true, fill: gray.lighten(60%)))
  })
  let arrow = line.with(stroke: 0.9pt + black, mark: (end: "stealth", fill: black))
  let branch(from, to, label) = {

```

```

    let corner = (from, "-|", to)
    arrow(from, corner, to)
    content((from, 50%, corner), label, anchor: "south", padding: 0.1)
  }

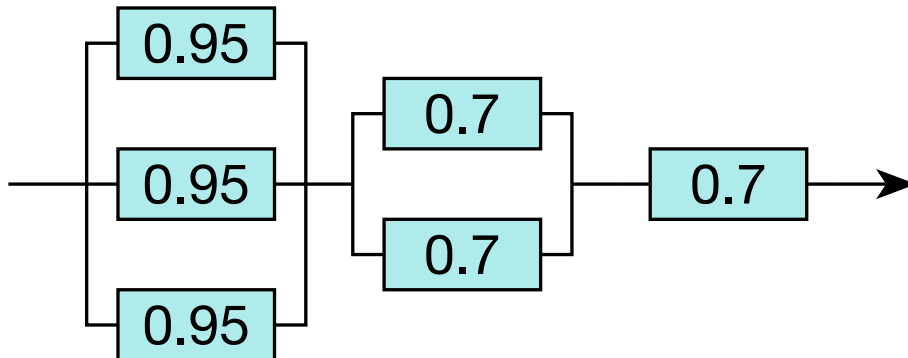
  let gap = 0.35
  step((0, 0), "plan", [Define sampling plan ($n_1$, $n_2$, $Ac_1$, $Ac_2$, $Re_1$)])
  step((rel: (0, -gap), to: "plan.south"), "ins1", [Inspect first sample $n_1$])
  decision((rel: (0, -gap), to: "ins1.south"), "d1", [Compare $d_1$, $Ac_1$ and $Re_1$])
  step((rel: (0, -2.5 * gap), to: "d1.south"), "ins2", [Inspect second sample $n_2$])
  decision((rel: (0, -gap), to: "ins2.south"), "d2", [Compare $d_1 + d_2$ and $Ac_2$])
  step((rel: (2 * gap, 0), to: "ins2.east"), "accept", [Accept lot],
    fill: eastern.lighten(60%), anchor: "west")
  step((rel: (-2 * gap, 0), to: "ins2.west"), "reject", [Reject lot],
    fill: orange.lighten(60%), anchor: "east")

  arrow("plan", "ins1")
  arrow("ins1", "d1")
  arrow("d1", "ins2")
  content(("d1.south", 50%, "ins2.north"), $Ac_1 < d_1 < Re_1$, anchor: "west", padding: 0.1)
  arrow("ins2", "d2")
  branch("d1.east", "accept.north", $d_1 \leq Ac_1$)
  branch("d1.west", "reject.north", $d_1 > Re_1$)
  branch("d2.east", "accept.south", $d_1 + d_2 \leq Ac_2$)
  branch("d2.west", "reject.south", $d_1 + d_2 > Ac_2$)
})

```

Series-parallel blocks

Redrawn from [circuitos_mistos1.pgf \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*. A data-driven loop; forks and joins with perpendicular coordinates; wires drawn behind the boxes with `on-layer`.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

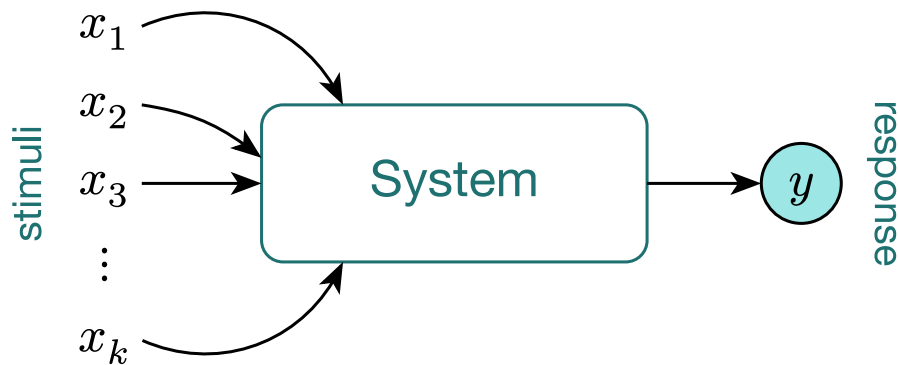
// Series-parallel reliability block diagram: each stage holds redundant
// components in parallel, and the stages are connected in series.
#cezt.canvas({
  import cetz.draw: *
  set-style(stroke: 0.6pt + black)

  let stages = ((0.95, 3), (0.7, 2), (0.7, 1)) // (reliability, count)
  let (dx, dy, w, h) = (1.7, 0.9, 1, 0.45)

  let prev = (rel: (-0.5, 0), to: "s0.west") // resolved when drawn, after "s0" exists
  for (i, (p, n)) in stages.enumerate() {
    let s = "s" + str(i)
    // The x-padding puts the group's west/east anchors on the fork and join buses.
    group(name: s, padding: (x: 0.2), {
      for k in range(n) {
        rect((i * dx, ((n - 1) / 2 - k) * dy), (rel: (w, h)), anchor: "center",
          fill: teal.lighten(60%), name: str(k))
        content(str(k), [#p])
      }
    })
    // Wires pass straight through the boxes, so draw them behind.
    on-layer(-1, {
      let (fork, join) = (s + ".west", s + ".east")
      line(prev, fork)
      for k in range(n) {
        let b = s + "." + str(k)
        line(fork, (fork, "|-", b), (join, "|-", b), join)
      }
    })
    prev = s + ".east"
  }
  line(prev, (rel: (0.5, 0)), mark: (end: "stealth", fill: black))
})
```

System with inputs and output

Redrawn from [reg_sistema.pgf \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*. Curved connectors ending at border anchors given as angles; a circle joined to a box; rotated captions.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

#let accent = teal.darken(45%)

#cetx.canvas({
  import cetx.draw: *
  set-style(
    stroke: 0.6pt + black,
    mark: (end: "stealth", fill: black, scale: 0.8),
    content: (padding: 0.08),
  )

  // Cubic curve leaving `a` and entering `b` at `angle` to the chord, like TikZ `bend left`.
  let bend(a, b, angle) = bezier(a, b, (a, 40%, angle, b), (b, 40%, -angle, a))

  rect((0, 0), (rel: (2.7, 1.1)), radius: 0.15, stroke: accent, name: "sys")
  content("sys", text(accent)[System])

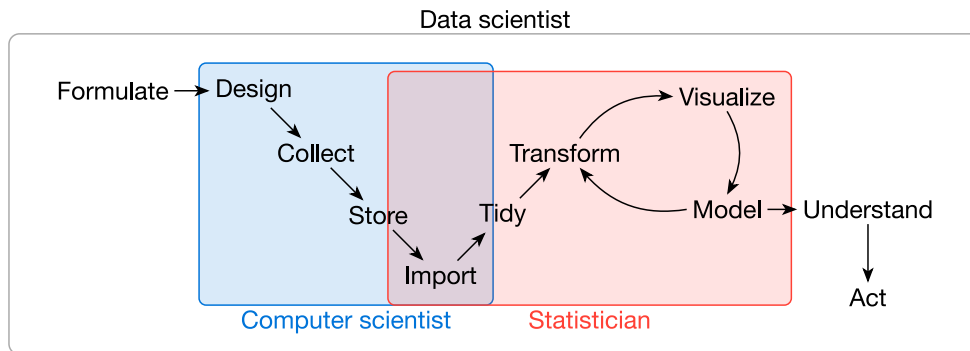
  let inputs = (x1: $x_1$, x2: $x_2$, x3: $x_3$, dots: $dots.v$, xk: $x_k$)
  for (i, (name, x)) in inputs.pairs().enumerate() {
    content((rel: (-1.1, (2 - i) * 0.55), to: "sys.west"), x, name: name)
  }
  bend("x1.east", (name: "sys", anchor: 120deg), 45deg)
  bend("x2.east", (name: "sys", anchor: 162deg), 15deg)
  line("x3", "sys")
  bend("xk.east", (name: "sys", anchor: 240deg), -45deg)

  circle((rel: (0.8, 0), to: "sys.east"), radius: 0.28, anchor: "west",
    fill: teal.lighten(50%), name: "y")
  content("y", $y$)
  line("sys", "y")

  set-style(content: (wrap: text.with(8pt, accent)))
  content((rel: (-0.1, 0), to: "x3.west"), [stimuli], angle: 90deg, anchor: "south")
  content((rel: (0.1, 0), to: "y.east"), [response], angle: -90deg, anchor: "south")
})
```

Overlapping groups

Redrawn from [data-science-workflow.pgf \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*. Translucent frames around groups of nodes with `rect-around`; a cycle of bent arrows; placement by polar offsets.



```

#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

#cetx.canvas({
  import cetx.draw: *
  set-style(
    stroke: 0.6pt + black,
    mark: (end: "stealth", fill: black, scale: 0.8),
    content: (padding: 0.08),
  )

  // Bare element names make `line` stop at both borders.
  let step(from, name, offset) = {
    content((rel: offset, to: from), name, name: name)
    line(from, name)
  }

  // Cubic curve leaving `a` and entering `b` at `angle` to the chord, like TikZ `bend left`.
  let bend(a, b, angle) = bezier(a, b, (a, 40%, angle, b), (b, 40%, -angle, a))

  content((0, 0), "Formulate", name: "Formulate")
  let prev = "Formulate"
  for (name, offset) in (
    ("Design", (0deg, 2)), ("Collect", (-45deg, 1.25)), ("Store", (-45deg, 1.25)),
    ("Import", (-45deg, 1.25)), ("Tidy", (45deg, 1.25)), ("Transform", (45deg, 1.25)),
  ) {
    step(prev, name, offset)
    prev = name
  }

  content((rel: (2.3, 0.8), to: "Transform"), "Visualize", name: "Visualize")
  content((rel: (2.3, -0.8), to: "Transform"), "Model", name: "Model")
  bend((name: "Transform", anchor: 45deg), "Visualize.west", 30deg)
  bend("Visualize.south", "Model.north", 30deg)
  bend("Model.west", (name: "Transform", anchor: -45deg), 30deg)

  step("Model", "Understand", (0deg, 2))
  step("Understand", "Act", (-90deg, 1.25))

  on-layer(-1, {
    for (members, hue, caption) in (
      (("Design", "Import"), blue, [Computer scientist]),
      (("Import", "Visualize"), red, [Statistician]),
    ) {
      rect-around(., members, padding: 0.15, radius: 0.1, name: "group",

```

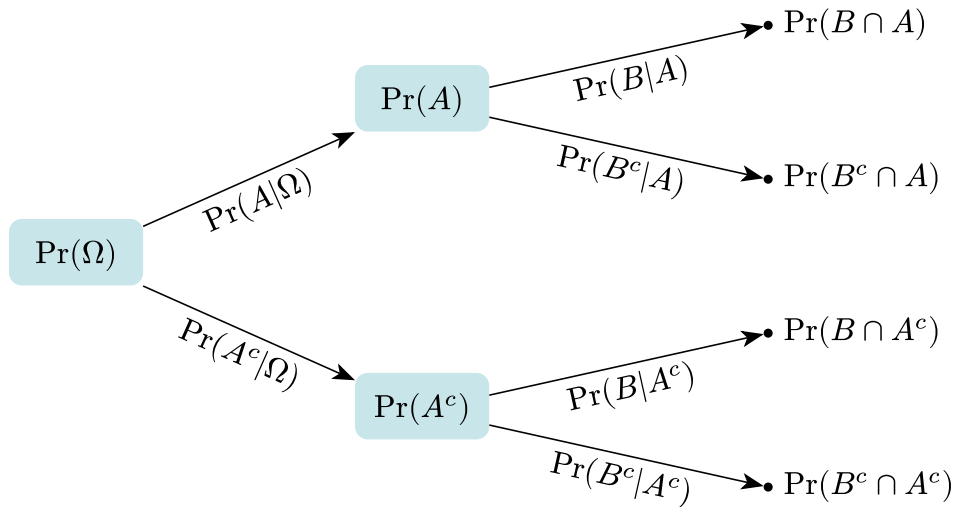
```

        stroke: hue, fill: hue.transparentize(85%)
    content("group.south", text(hue, caption), anchor: "north")
}
rect-around("Formulate", "Import", "Understand", "Act", padding: 0.6, radius: 0.15,
    stroke: gray, name: "all")
content("all.north", [Data scientist], anchor: "south")
})
})

```

Probability tree

Redrawn from [probtreescheme.png](#) in Walmes M. Zeviani's *Tikz Gallery*. `cetz.tree` growing to the right; custom node and edge callbacks; labels along the edges.



```

#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

#cetz.canvas({
  import cetz.draw: *
  set-style(
    stroke: .5pt + black,
    mark: (end: "stealth", fill: black),
    content: (padding: .12),
  )

  let draw-node(n) = if n.depth < 2 {
    let bag = box.with(fill: eastern.lighten(75%), radius: 4pt, inset: 7pt, width: 4.2em)
    content((0, 0), padding: 0, bag(align(center, $Pr(#n.content)$)))
  } else {
    circle((0, 0), radius: .05, fill: black, stroke: none)
  }

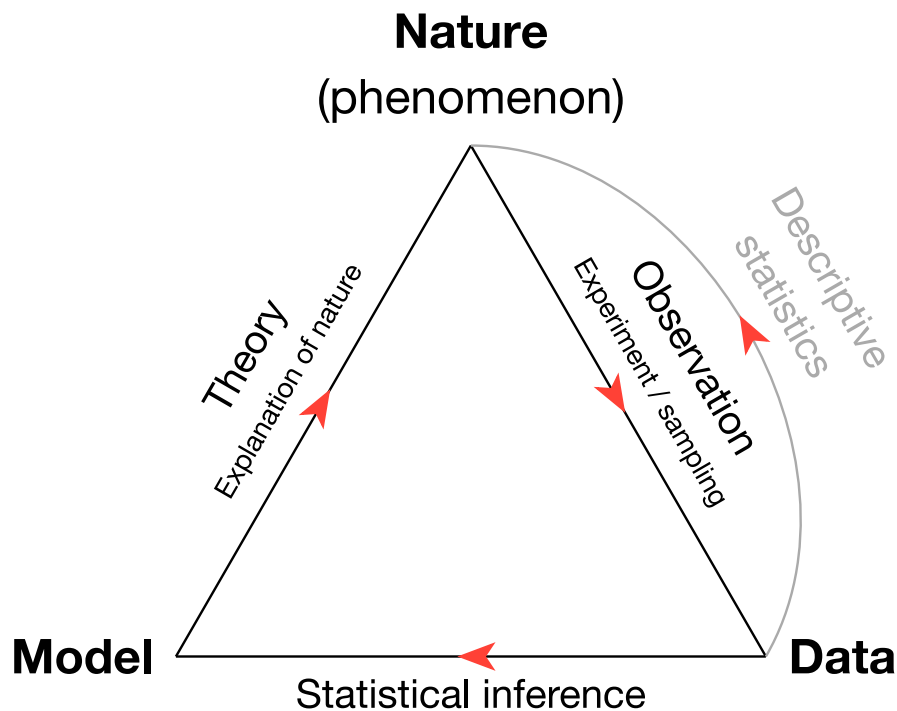
  let draw-edge(parent, child) = {
    let (a, b) = (parent.content, child.content)
    line(parent.group-name, child.group-name, name: "e")
    content("e.mid", $Pr(#b|#a)$, angle: "e.end", anchor: "north")
    // Only the edge callback sees both events, so it also labels the leaf.
    if child.depth == 2 {
      content(child.group-name + ".east", $Pr(#b inter #a)$, anchor: "west")
    }
  }

  // With direction "right", children are listed bottom to top.
  cetz.tree.tree(
    ($0\Omega$, ($A^c$, $B^c$, $B$), ($A$, $B^c$, $B$)),
    direction: "right", grow: 3, spread: 1.6,
    draw-node: draw-node, draw-edge: draw-edge,
  )
})

```

Cycle on a triangle

Redrawn from [knowledge-cycle.pgfr \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*. Polar coordinates; arrowheads in the middle of each edge; text rotated along edges; a Bézier curve with polar control points.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

#let note(title, sub) = align(center)[#title \ #text(size: 7pt, sub)]

#cetz.canvas({
  import cetz.draw: *
  set-style(
    stroke: .6pt + black,
    // Mid-path arrowhead; shorten-to: none keeps CetZ from cutting the path at the mark.
    mark: (end: "stealth", pos: 50%, anchor: "center", shorten-to: none,
      scale: 1.4, fill: red, stroke: red),
    content: (padding: .2),
  )

  anchor("nature", (90deg, 3))
  anchor("model", (210deg, 3))
  anchor("data", (330deg, 3))
  content("nature", text(12pt, align(center)[*Nature* \ (phenomenon)]), anchor: "south")
  content("model", text(12pt)[*Model*], anchor: "east")
  content("data", text(12pt)[*Data*], anchor: "west")

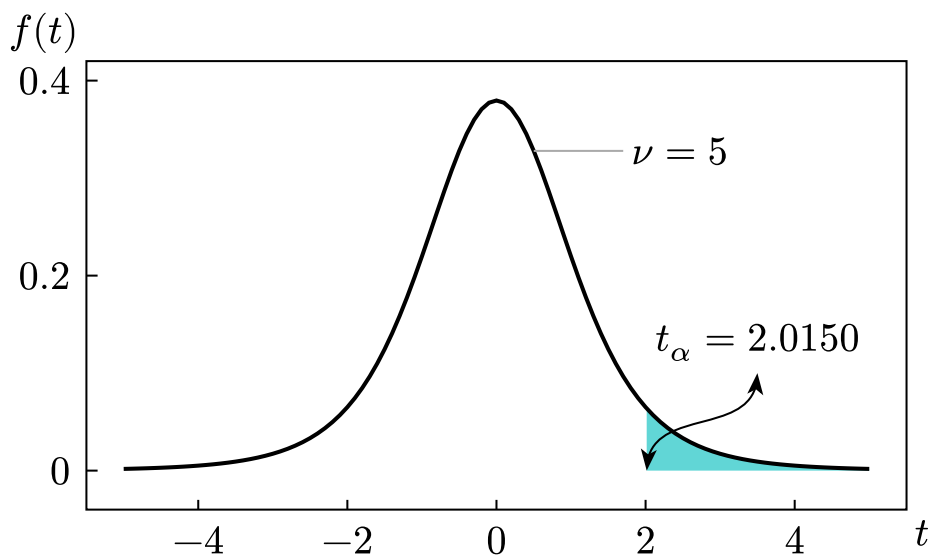
  let edge(a, b, label) = {
    line(a, b, name: "e")
    content("e.mid", label, angle: "e.end", anchor: "south")
  }

  edge("model", "nature", note[Theory][Explanation of nature])
  edge("nature", "data", note[Observation][Experiment / sampling])
  line("data", "model", name: "inference")
  content("inference.mid", [Statistical inference], anchor: "north")
})
```

```
bezier("data", "nature", (rel: (60deg, 2), to: "data"), (rel: (0deg, 2), to: "nature"),  
  name: "arc", stroke: gray, mark: (fill: red, stroke: red))  
content("arc.mid", angle: -60deg, anchor: "south",  
  text(fill: gray, align(center)[Descriptive \ statistics]))  
})
```

Density with a shaded tail

Redrawn from [dist_t_quantil.pgf \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*. A plot in data units via `scale`; a sampled curve; a filled tail; tick loops; a curved callout.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

#cezt.canvas({
  import cetz.draw: *
  set-style(stroke: (paint: black, thickness: .5pt), mark: (fill: black, scale: .7))
  // Data coordinates (t, f(t)); text, strokes and marks keep their size.
  scale(x: .65, y: 8.5)

  // Student t density for ν = 5: Γ(3) / (√(5π) Γ(5/2)) · (1 + t²/5)^(-3).
  let pdf(t) = 8 / (3 * calc.pi * calc.sqrt(5)) * calc.pow(1 + t * t / 5, -3)
  let curve(a, b) = range(101).map(i => a + (b - a) * i / 100).map(t => (t, pdf(t)))
  let q = 2.015

  line((q, 0), ..curve(q, 5), (5, 0), close: true, stroke: none,
    fill: teal.lighten(20%))
  line(..curve(-5, 5), stroke: (thickness: 1pt))

  rect((-5.5, -.04), (5.5, .42), name: "frame")
  for x in (-4, -2, 0, 2, 4) {
    let p = ((x, 0), "|-", "frame.south")
    line(p, (rel: (0, .01)))
    content(p, $#x$, anchor: "north", padding: 4pt)
  }
  for y in (0, .2, .4) {
    let p = ("frame.west", "|-", (0, y))
    line(p, (rel: (.15, 0)))
    content(p, $#y$, anchor: "east", padding: 4pt)
  }
  content("frame.south-east", $t$, anchor: "north-west", padding: 2pt)
  content("frame.north-west", $f(t)$, anchor: "south-east", padding: 2pt)

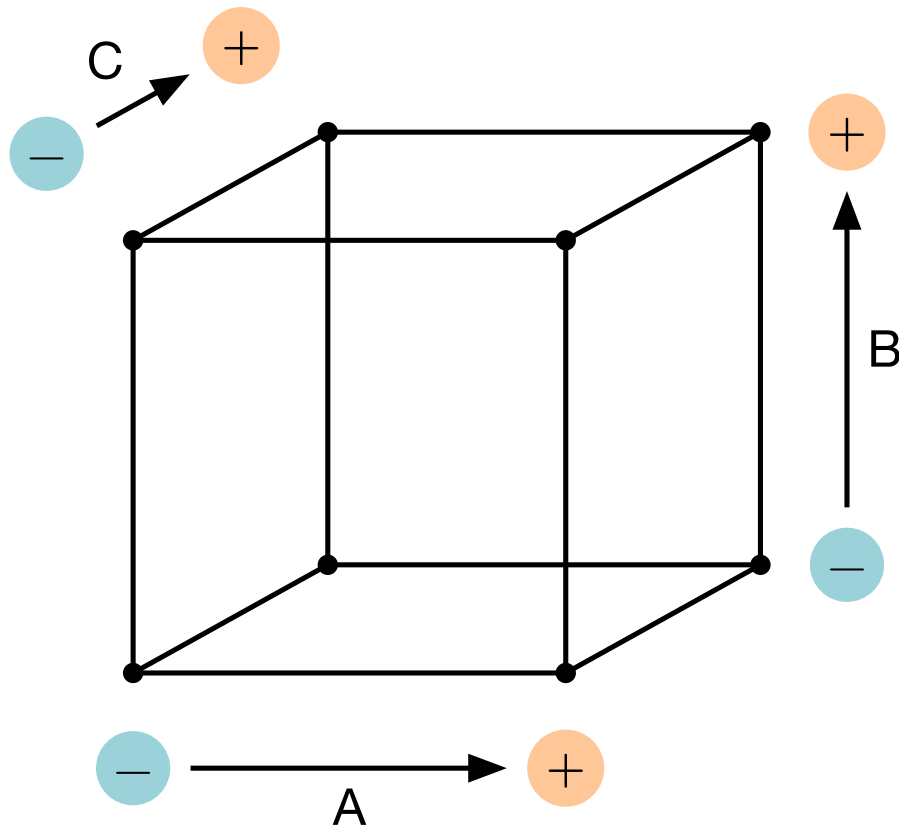
  line(.5, pdf(.5), (rel: (1.2, 0)), stroke: gray, name: "leader")
  content("leader.end", $nu = 5$, anchor: "west", padding: 2pt)

  bezier((q, 0), (3.5, .1), (q, .07), (3.5, .03),
```

```
mark: (start: "stealth", end: "stealth"), name: "callout")
content("callout.end", $t_alpha = 2.0150$, anchor: "south", padding: 3pt)
})
```

Factorial-design cube

Redrawn from [cubos-fatoriais-2a3.png](#) (png) in Walmes M. Zeviani's *Tikz Gallery*. An oblique 3D view from the canvas's z vector; a loop over vertices; framed circle markers joined by arrows.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

// 2^3 factorial design: each vertex is one run, each axis one factor at levels -/+.
// Oblique projection: the canvas draws the z unit vector as (.45, .25).
#cetz.canvas(length: 3cm, z: (.45, .25), {
  import cetz.draw: *
  set-style(stroke: black, mark: (fill: black))

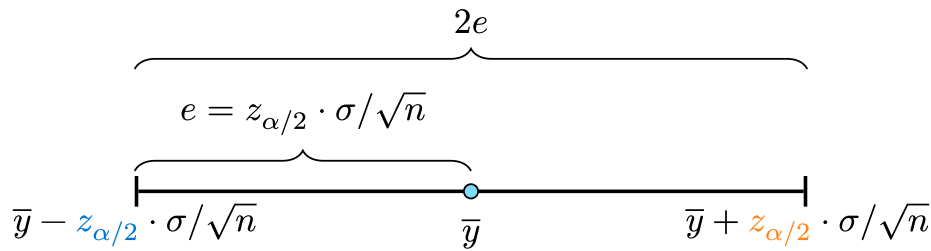
  for x in (0, 1) { for y in (0, 1) { for z in (0, 1) {
    let p = (x, y, z)
    if x == 0 { line(p, (rel: (1, 0, 0))) }
    if y == 0 { line(p, (rel: (0, 1, 0))) }
    if z == 0 { line(p, (rel: (0, 0, 1))) }
    circle(p, radius: 2pt, fill: black, stroke: none)
  } } }

  // Level discs beside the low and high vertex; a line between named
  // elements stops at their borders.
  set-style(content: (frame: "circle", stroke: none, padding: 1.5pt))
  let factor(id, low, high, offset, anchor) = {
    content((rel: offset, to: low), $-$, fill: eastern.transparentize(55%), name: id + "-")
    content((rel: offset, to: high), $+$, fill: orange.transparentize(55%), name: id + "+")
    // The invisible start mark makes `offset` leave a gap at both ends.
    line(id + "-", id + "+", mark: (start: (symbol: "|", stroke: none), end: ">", offset: 4pt))
    content((id + "-", 50%, id + "+"), id, anchor: anchor, frame: none, padding: 4pt)
```

```
}  
factor("A", (0, 0, 0), (1, 0, 0), (0, -.22), "north")  
factor("B", (1, 0, 1), (1, 1, 1), (.2, 0), "west")  
factor("C", (0, 1, 0), (0, 1, 1), (-.2, .2), "south-east")  
})
```

Interval with braces

Redrawn from [confidence-interval-mean-components.pgf \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*. flat-brace with labels; interval end marks; colored parts of math labels.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

// Margin of error; `fill` colors the z.
#let half(fill) = $#text(fill: fill, $z_{\alpha/2}) dot sigma \ / sqrt(n)$

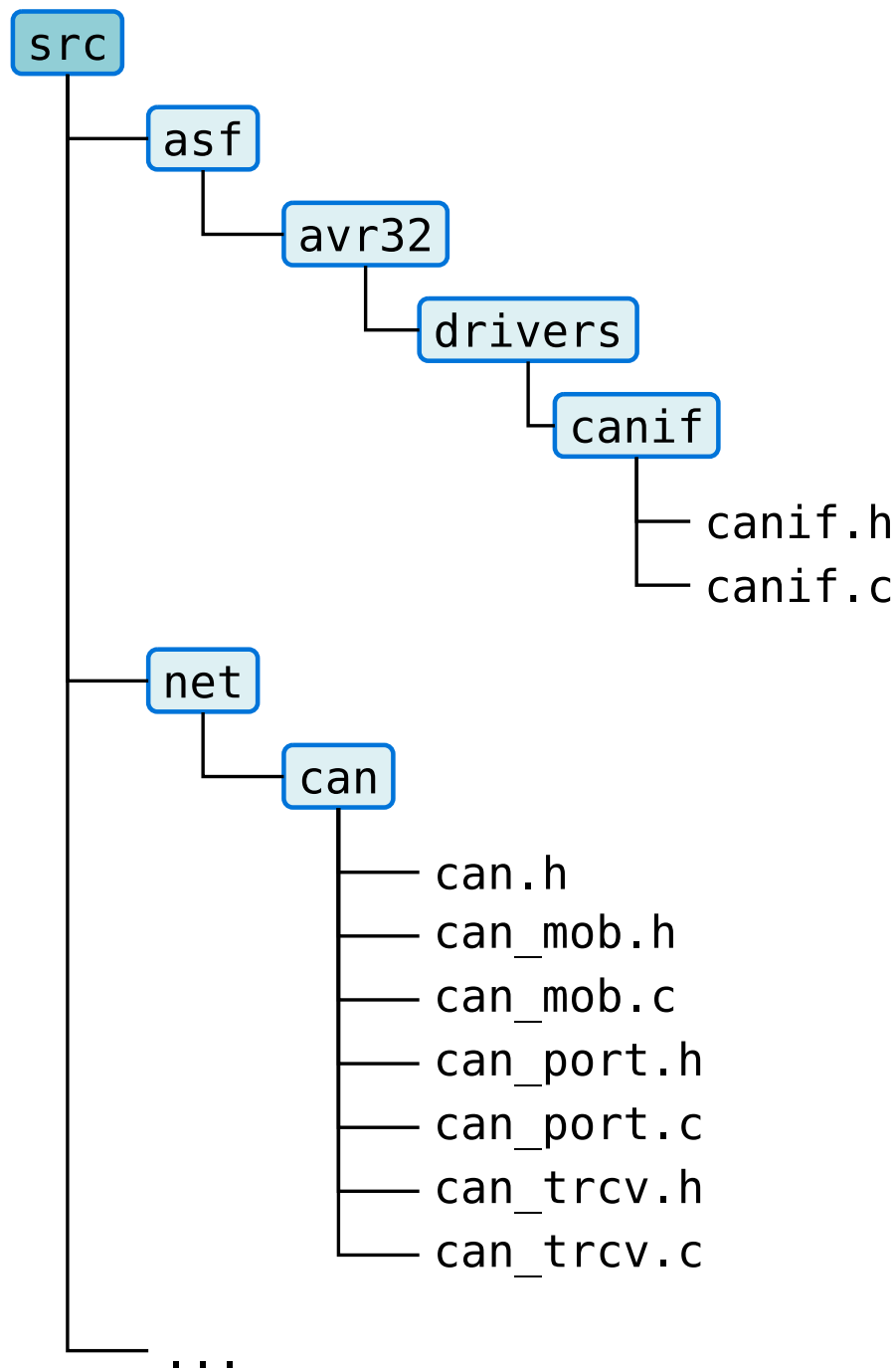
#set canvas({
  import cetz.draw: *
  import cetz.decorations: flat-brace
  set-style(
    stroke: (paint: black, thickness: .5pt),
    content: (padding: .15),
    flat-brace: (amplitude: .2, content-offset: 0),
  )

  line((-3.3, 0), (3.3, 0), name: "ci", stroke: (thickness: 1pt), mark: (symbol: "|",
width: .3))
  circle("ci.50%", radius: .07, fill: aqua)
  content("ci.50%", $overline(y)$, anchor: "north", padding: .25)
  content("ci.start", $overline(y) - #half(blue)$, anchor: "north")
  content("ci.end", $overline(y) + #half(orange)$, anchor: "north")

  flat-brace((rel: (0, .2), to: "ci.start"), (rel: (0, .2), to: "ci.50%"), name: "e")
  content("e.content", $e = #half(black)$, anchor: "south")
  flat-brace((rel: (0, 1.2), to: "ci.start"), (rel: (0, 1.2), to: "ci.end"), name: "2e")
  content("2e.content", $2e$, anchor: "south")
})
```

Directory tree

Redrawn from [file_tree.pgf \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*. A tree drawn from a flat data list; elbow connectors from each parent's south to each child's west.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

// (depth, name) in `tree` order; an entry is a directory if the next one is deeper.
#let entries = (
  (0, "src"),
  (1, "asf"), (2, "avr32"), (3, "drivers"), (4, "canif"),
```

```

(5, "canif.h"), (5, "canif.c"),
(1, "net"), (2, "can"),
..("can.h", "can_mob.h", "can_mob.c", "can_port.h", "can_port.c", "can_trcv.h", "can_trcv.c")
.map(file => (3, file)),
(1, "..."),
)

#cetz.canvas({
  import cetz.draw: *
  set-style(stroke: (paint: black, thickness: .6pt))

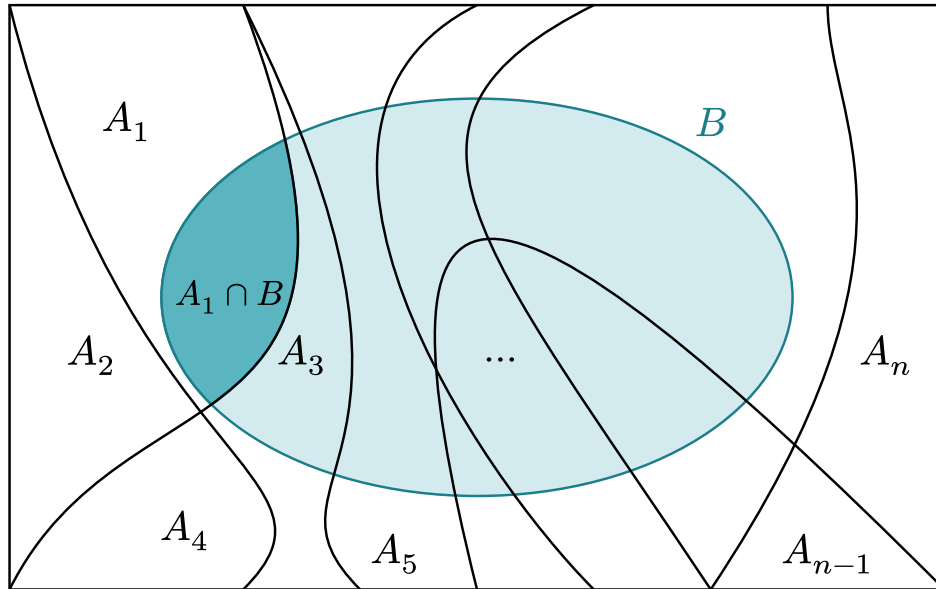
  let y = 0
  let prev-depth = 0
  for (i, (depth, name)) in entries.enumerate() {
    let dir = entries.at(i + 1, default: (0,)).at(0) > depth
    y -= if depth == prev-depth { .4 } else { .6 }
    prev-depth = depth

    // Reused names: "d2" is the latest depth-2 node, i.e. the parent of each new depth-3 node.
    let node = "d" + str(depth)
    content((depth * .85, y), name: node, anchor: "west", box(
      raw(name), inset: 2.5pt, radius: 2pt,
      fill: if depth == 0 { eastern.lighten(50%) } else if dir { eastern.lighten(85%) },
      stroke: if dir { blue + .8pt },
    ))
    if depth > 0 {
      let parent = "d" + str(depth - 1) + ".south"
      line(parent, (parent, "|-", node + ".west"), node + ".west")
    }
  }
})

```

Venn partition

Redrawn from [diagrama_venn4.pg](#) (png) in Walmes M. Zeviani's *Tikz Gallery*. `boolean(op: "intersection")` of a closed merge-path and a named ellipse, in place of clipping; loops for the partition curves and labels.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

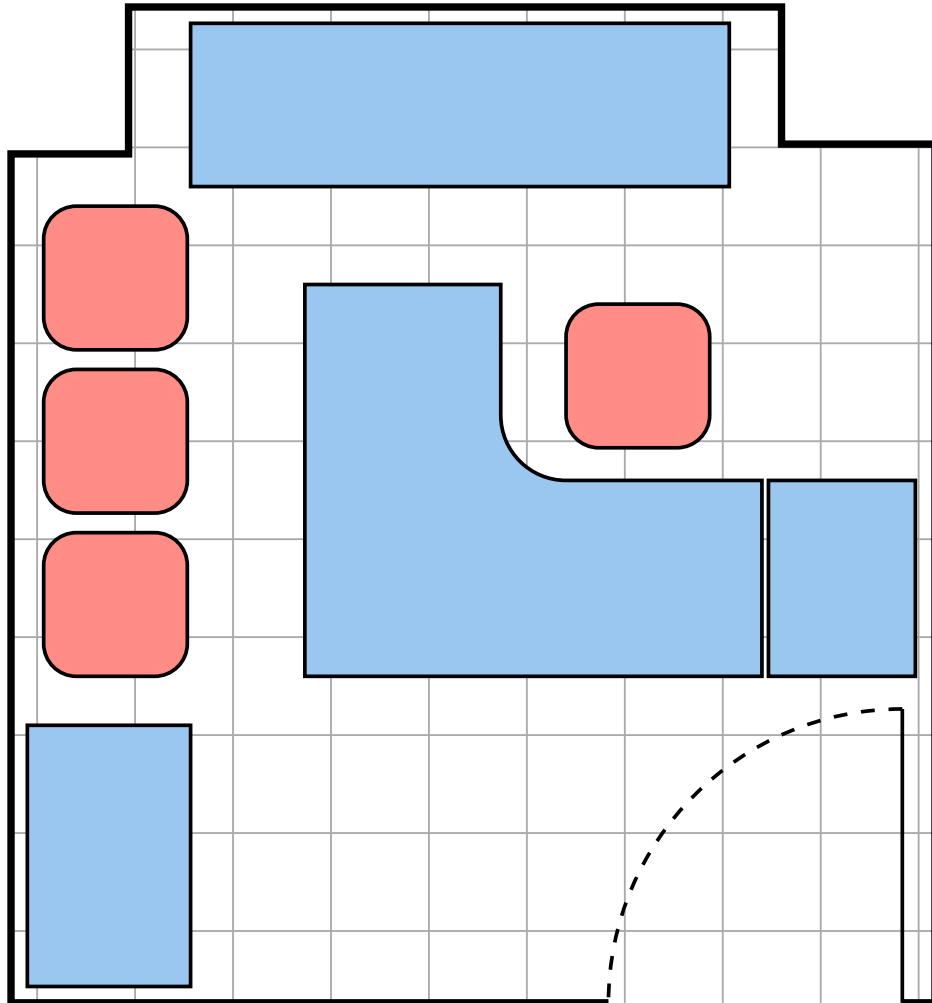
#cetx.canvas({
  import cetx.draw: *
  set-style(stroke: 0.6pt + black)
  let hue = eastern
  let border-a1 = bezier.with((0, 0), (2, 5), (1, 2), (3.5, 1))

  circle((4, 2.5), radius: (2.7, 1.7), fill: hue.lighten(80%), stroke: hue.darken(20%), name:
  "b")
  // The black border of A1 is drawn over the stroke of the intersection.
  boolean(merge-path(close: true, { border-a1(); line((0, 5)) }, "b", op: "intersection",
    fill: hue.lighten(25%), stroke: hue.darken(20%))
  rect((0, 0), (8, 5))
  border-a1()
  // Boundaries between the remaining events, as (start, end, control 1, control 2).
  for (a, b, c1, c2) in (
    ((0, 5), (2, 0), (1, 1), (3, 1)),
    ((3, 0), (2, 5), (2, 1), (4, 1)),
    ((4, 5), (5, 0), (2, 4), (4, 1)),
    ((4, 0), (8, 0), (3, 4), (4, 4)),
    ((5, 5), (6, 0), (3, 4), (4, 3)),
    ((7, 5), (6, 0), (7, 4), (8, 3)),
  ) { bezier(a, b, c1, c2) }

  content((1.88, 2.5), text(8.5pt, $A_1 inter B$))
  content((6, 4), text(hue.darken(20%), $B$))
  for (pos, label) in (
    ((1, 4), $A_1$), ((0.7, 2), $A_2$), ((2.5, 2), $A_3$), ((1.5, 0.5), $A_4$),
    ((3.3, 0.3), $A_5$), ((4.2, 2), $dots.c$), ((7, 0.3), $A_{n-1}$), ((7.5, 2), $A_n$),
  ) { content(pos, label) }
})
```

Floor plan

Redrawn from [floor-plan.pgf \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*. Walls from a list of `rel:` moves; `boolean(op: "difference")` hides the grid outside the room; a door swing `arc` from an anchor; furniture as rotated groups with a scoped style.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

#cetz.canvas(length: 2.5mm, {
  import cetz.draw: *
  set-style(stroke: 0.8pt + black)

  // Walls run counterclockwise from the door hinge to the far door jamb.
  let hinge = (27.3, 0)
  let walls = (hinge, (rel: (1, 0)), (rel: (0, 26.3)), (rel: (-4.7, 0)), (rel: (0, 4.2)),
    (rel: (-20, 0)), (rel: (0, -4.5)), (rel: (-3.6, 0)), (rel: (0, -26)), (rel: (18.3, 0)))
  let bounds = ((0, 0), (28.3, 30.5))

  grid(..bounds, step: 3, shift: (0.8, 2.2), stroke: 0.4pt + gray)
  boolean(rect(..bounds), line(..walls, close: true), op: "difference", fill: white, stroke:
none)
  line(..walls, stroke: 1.6pt)

  line(hinge, (rel: (0, 9)), name: "leaf")
}
```

```

arc("leaf.end", start: 90deg, delta: 90deg, radius: 9, stroke: (dash: "dashed"))

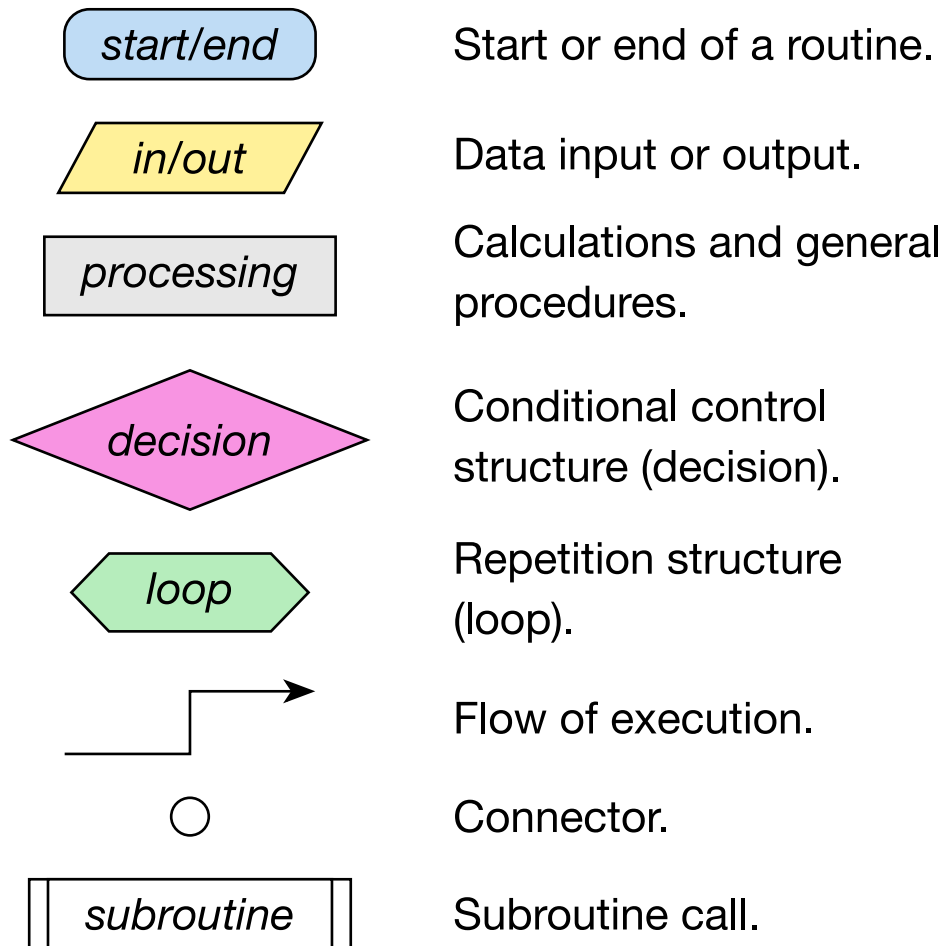
let piece(origin, angle, body) = group({
  set-style(fill: blue.lighten(60%)); translate(origin); rotate(angle); body })
piece((0.5, 8.5), -90deg, rect((0, 0), (8, 5))) // bookcase
piece((22, 25), 90deg, rect((0, 0), (5, 16.5))) // low cupboard
piece((9, 22), -90deg, {
  // desk and drawer unit
  merge-path(close: true, {
    line((0, 0), (rel: (12, 0)), (rel: (0, 14)), (rel: (-6, 0)), (rel: (0, -6)))
    arc((0, 0), start: 0deg, delta: -90deg, radius: 2)
    line((0, 0), (0, 6))
  })
  rect((6, 14.2), (rel: (6, 4.5)))
})

for pos in ((17, 17), (1, 20), (1, 15), (1, 10)) {
  rect(pos, (rel: (4.4, 4.4)), radius: 1, fill: red.lighten(40%))
}
})

```

Flowchart shapes

Redrawn from [fluxograms.pgf \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*, which is based on [an answer by Claudio Fiandrino](#) on TeX Stack Exchange (CC BY-SA 3.0). Shapes as small functions that draw their outline from the label's anchors (rect-around, parallelogram, hexagon, diamond); rows from a data list, stacked with "|" coordinates.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)
// Based on https://tex.stackexchange.com/a/87956 by Claudio Fiandrino, CC BY-SA 3.0.

#cetx.canvas({
  import cetx.draw: *
  set-style(stroke: 0.6pt + black)

  // A shape is its label "t" plus an outline drawn around the label's anchors.
  let node(body, outline, pad-x: 0.3) = pos => {
    content(pos, emph(body), name: "t", padding: (x: pad-x, y: 0.15))
    on-layer(-1, outline)
  }
  let chamfer = 0.3
  let rows = (
    (node([start/end], rect-around("t", radius: 0.2, fill: blue.lighten(75%))),
      [Start or end of a routine.]),
    (node([in/out], line((rel: (-chamfer, 0), to: "t.south-west"), "t.south-east",
      (rel: (chamfer, 0), to: "t.north-east"), "t.north-west", close: true,
      fill: yellow.lighten(60%))),

```

```

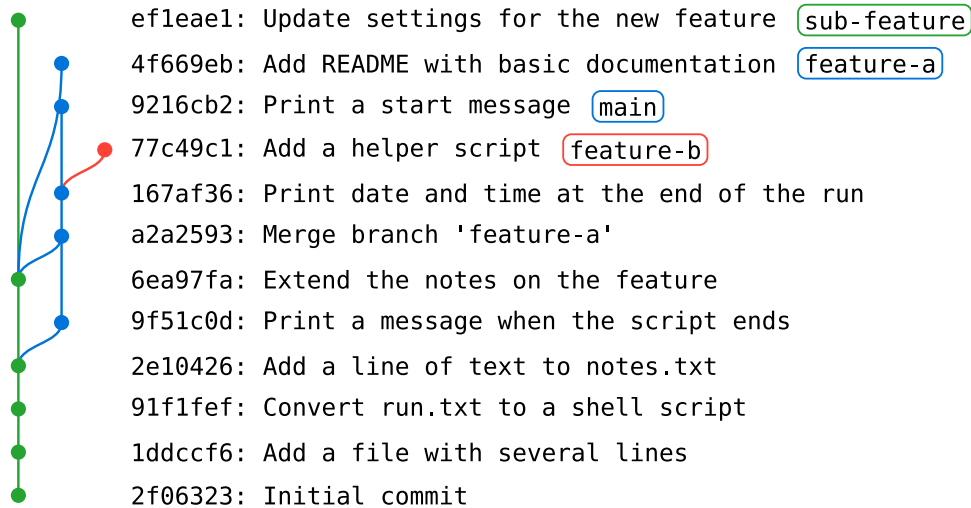
[Data input or output.]),
(node([processing], rect-around("t", fill: silver.lighten(30%))),
[Calculations and general \ procedures.]),
// Vertices at twice the label's half-size put the label's corners on the edges.
(node([decision], line(..("north", "east", "south", "west").map(a => ("t", 200%, "t." + a)),
close: true, fill: fuchsia.lighten(55%)), pad-x: 0.05),
[Conditional control \ structure (decision).]),
(node([loop], line((rel: (-chamfer, 0), to: "t.west"), "t.north-west", "t.north-east",
(rel: (chamfer, 0), to: "t.east"), "t.south-east", "t.south-west", close: true,
fill: green.lighten(65%))),
[Repetition structure \ (loop).]),
(pos => line((rel: (-1, -0.25), to: pos), (rel: (1, 0)), (rel: (0, 0.5)), (rel: (1, 0)),
mark: (end: "stealth", fill: black)),
[Flow of execution.]),
(pos => circle(pos, radius: 0.15), [Connector.]),
// The side bars sit outside the label box, as in TikZ's `predefined process`.
(node([subroutine], {
rect-around("t", padding: (x: 0.15))
rect-around("t", fill: white)
}), [Subroutine call.]),
)

// Shapes share a centre line; each row starts below the lower of the previous shape and text.
let text-x = 2.1
for (i, (shape, desc)) in rows.enumerate() {
let top = if i == 0 { (0, 0) } else { ((0, 0), "|-", (rel: (0, -0.35), to: "row" + str(i -
1) + ".south")) }
group(name: "row" + str(i), {
group(name: "shape", anchor: "north", shape(top))
content(((text-x, 0), "|-", "shape"), desc, anchor: "west")
})
}
})

```

Commit graph

Redrawn from [git_log.png](#) in Walmes M. Zeviani's *Tikz Gallery*, which is based on [an answer by ph0t0nix](#) on TeX Stack Exchange (CC BY-SA 3.0). Layout from a data list of commits, lanes and parents; bezier edges that leave and enter vertically; messages and branch tags placed with perpendicular and relative coordinates.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)
// Based on https://tex.stackexchange.com/a/262163 by ph0t0nix, CC BY-SA 3.0.

// Newest first, as in `git log --graph`: (hash, lane, parents, message, branch tag).
#let commits = (
  ("ef1eae1", 0, ("6ea97fa"), "Update settings for the new feature", "sub-feature"),
  ("4f669eb", 1, ("6ea97fa"), "Add README with basic documentation", "feature-a"),
  ("9216cb2", 1, ("167af36"), "Print a start message", "main"),
  ("77c49c1", 2, ("167af36"), "Add a helper script", "feature-b"),
  ("167af36", 1, ("a2a2593"), "Print date and time at the end of the run", none),
  ("a2a2593", 1, ("9f51c0d", "6ea97fa"), "Merge branch 'feature-a'", none),
  ("6ea97fa", 0, ("2e10426"), "Extend the notes on the feature", none),
  ("9f51c0d", 1, ("2e10426"), "Print a message when the script ends", none),
  ("2e10426", 0, ("91f1fef"), "Add a line of text to notes.txt", none),
  ("91f1fef", 0, ("1ddccf6"), "Convert run.txt to a shell script", none),
  ("1ddccf6", 0, ("2f06323"), "Add a file with several lines", none),
  ("2f06323", 0, (), "Initial commit", none),
)
#let lanes = (green.darken(20%), blue, red)

#cetx.canvas({
  import cetx.draw: *

  for (row, (id, lane, _, msg, tag)) in commits.enumerate() {
    let hue = lanes.at(lane)
    circle((lane * 0.5, -row * 0.5), radius: 0.07, fill: hue, stroke: hue, name: id)
    // Messages share a column right of the widest lane, so no edge runs through them.
    content(((1.3, 0), "|-", id), raw(id + ": " + msg), anchor: "west", name: id + "-msg")
    if tag != none {
      content((rel: (0.25, 0), to: id + "-msg.east"), anchor: "west",
        box(raw(tag), stroke: 0.6pt + hue, radius: 3pt, inset: 2pt))
    }
  }
})

// Each edge leaves the parent and enters the child vertically, in the child's lane colour.
```

```
on-layer(-1, for (id, lane, parents, ..) in commits {  
  for parent in parents {  
    let mid = (parent, 50%, id)  
    bezier(parent, id, (parent, "|-", mid), (id, "|-", mid), stroke: 0.8pt + lanes.at(lane))  
  }  
})  
})
```

Mind map

Redrawn from [mindmapImovel.pgf \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*. Children at polar positions from a loop; tapered connectors as a merge-path of two bezier curves; colors from `color.mix`; `gradient.linear` fills behind the circles.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

#cetx.canvas({
  import cetx.draw: *
  set-style(stroke: none)

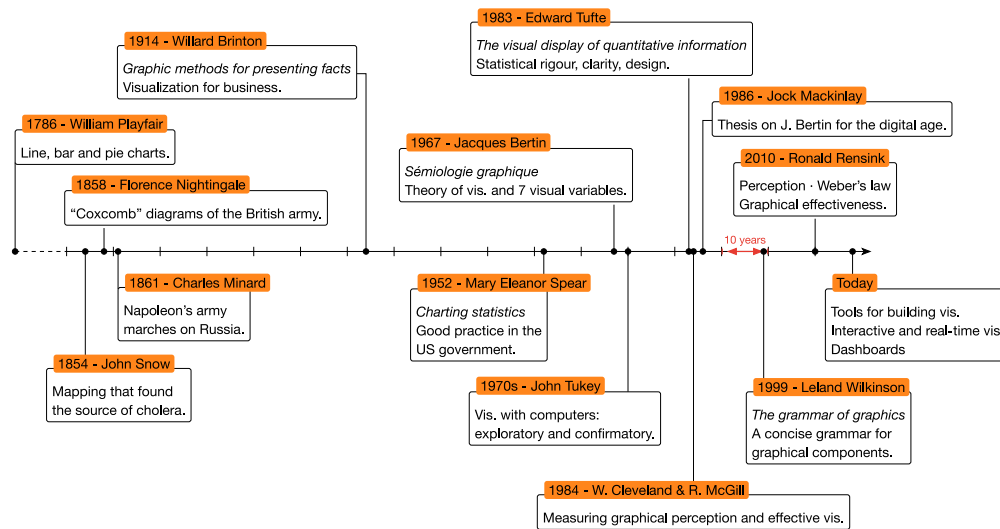
  let (R, r, d) = (1.7, 1.1, 4.4)
  let root = teal.darken(40%)
  let concept(pos, radius, fill, body, ink: white) = {
    circle(pos, radius: radius, fill: fill)
    content(pos, align(center, text(ink, body)))
  }
  // Tapered bar from the root towards `a` with waist 2w; Typst gradient angles run clockwise.
  let bar(a, fill, w: 0.08) = {
    let side(s) = (
      (a + s * 20deg, R),
      (rel: (a + 180deg - s * 30deg, r), to: (a, d)),
      (rel: (a + s * 90deg, w), to: (a, R + 0.7)),
      (rel: (a + s * 90deg, w), to: (a, d - r - 0.6)),
    )
    let (p, q, c1, c2) = side(-1)
    merge-path(close: true, fill: gradient.linear(root, fill, angle: -a), {
      bezier(..side(1))
      bezier(q, p, c2, c1)
    })
  }

  concept((0, 0), R, root, text(13pt)[Sale price \ of a property])
  for (i, (hue, ink, body)) in (
    (white, black, [Area (m²), \ Bedrooms, \ Garage, \ Pool]),
    (blue, white, [Age, \ Condition, \ Finish]),
    (orange, black, [Location, \ Access, \ Security]),
    (purple, white, [Economic \ indicators]),
  ).enumerate() {
```

```
let a = 180deg - i * 60deg
let fill = color.mix((root, 60%), (hue, 40%))
on-layer(-1, bar(a, fill))
concept((a, d), r, fill, ink: ink, body)
}
})
```

Timeline

Redrawn from [timeline_dataviz.pgf \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*. A data-driven callout helper placed by an angle anchor; leaders with "|-" coordinates; tag tabs relative to a box corner; a year-to-position mapping function.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

#cetx.canvas(length: 1.2cm, {
  import cetx.draw: *
  set-style(stroke: 0.6pt + black)

  // One unit per decade from 1850; the dashed stub stands in for the years back to 1786.
  let x(year) = calc.max((year - 1850) / 10, -1.1)
  line((-1.1, 0), (0, 0), stroke: (dash: "dashed"))
  line((0, 0), (17.2, 0), mark: (end: "stealth", fill: black))
  for i in range(17) { line((i, -0.1), (i, 0.1)) }
  let rule = red.darken(10%)
  line((14, 0), (15, 0), name: "scale", stroke: rule + 0.9pt,
    mark: (symbol: ("|", ">"), fill: rule))
  content("scale.mid", text(8pt, rule)[10 years], anchor: "south", padding: 0.12)

  // The box's border point at angle `at` lies at `offset` from the event.
  let callout(year, offset, at, title, body) = {
    let p = (x(year), 0)
    let a = (name: "box", anchor: at)
    content((rel: offset, to: p), name: "box", anchor: at,
      box(fill: white, stroke: 0.6pt, radius: 2pt, inset: (x: 4pt, top: 13pt, bottom: 5pt),
        body))
    content((rel: (0.15, 0.1), to: "box.north-west"), anchor: "west",
      box(fill: orange, radius: 2pt, inset: 3pt, title))
    on-layer(-1, line(p, (p, "|-", a), a))
    circle(p, radius: 2pt, fill: black)
  }

  for e in (
    (1786, (0, 1.9), -170deg, [1786 - William Playfair],
      [Line, bar and pie charts.]),
    (1854, (0, -2.5), 140deg, [1854 - John Snow],
      [Mapping that found \ the source of cholera.]),
    (1858, (0, 0.5), -168.5deg, [1858 - Florence Nightingale],
      ["Coxcomb" diagrams of the British army.]),
    (1861, (0, -0.8), 158deg, [1861 - Charles Minard],
```

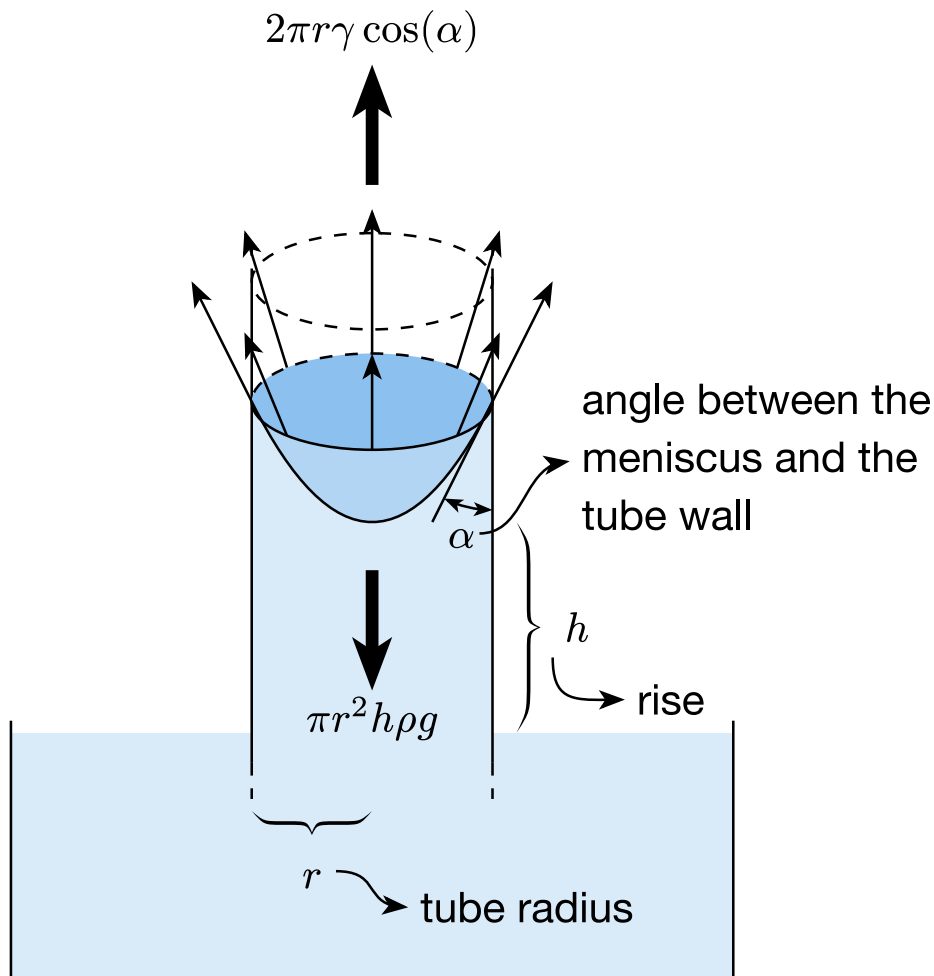
```

[Napoleon's army \ marches on Russia.]),
(1914, (-0.2, 3.8), 0deg, [1914 - Willard Brinton],
[_Graphic methods for presenting facts_ \ Visualization for business.]),
(1952, (0, -0.8), 30deg, [1952 - Mary Eleanor Spear],
[_Charting statistics_ \ Good practice in the \ US government.]),
(1967, (0, 1.0), -16deg, [1967 - Jacques Bertin],
[_Sémiologie graphique_ \ Theory of vis. and 7 visual variables.]),
(1970, (0, -3.0), 24deg, [1970s - John Tukey],
[Vis. with computers: \ exploratory and confirmatory.]),
(1983, (0, 3.7), -20deg, [1983 - Edward Tufte],
[_The visual display of quantitative information_ \ Statistical rigour, clarity,
design.]),
(1984, (0, -5.2), 90deg, [1984 - W. Cleveland & R. McGill],
[Measuring graphical perception and effective vis.]),
(1986, (0.2, 2.8), 180deg, [1986 - Jock Mackinlay],
[Thesis on J. Bertin for the digital age.]),
(1999, (0, -3.0), 150deg, [1999 - Leland Wilkinson],
[_The grammar of graphics_ \ A concise grammar for \ graphical components.]),
(2010, (0, 0.7), -90deg, [2010 - Ronald Rensink],
[Perception · Weber's law \ Graphical effectiveness.]),
(2018, (0, -0.8), 150deg, [Today],
[Tools for building vis. \ Interactive and real-time vis. \ Dashboards]),
) { callout(..e) }
})

```

Capillary rise

Redrawn from [bouma.pgf \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*. A pseudo-3D tube from elliptical arcs; arrows at elliptical polar coordinates; `cetz.angle.angle` for the contact angle; brace for dimensions; curved callouts.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

#cetz.canvas({
  import cetz.draw: *
  import cetz.decorations: brace
  set-style(stroke: .6pt + black, mark: (fill: black, scale: .8), brace: (amplitude: .2,
content-offset: .1),
    content: (padding: .1))
  let arrow = line.with(mark: (end: "stealth"))
  let callout(from, to, body, out: 0deg) = {
    bezier(from, to, (rel: (out, .4), to: from), (rel: (-.4, 0), to: to), mark: (end:
"stealth"))
    content(to, body, anchor: "west")
  }
  let water = blue.lighten(85%)
  // Tube radius and its foreshortened depth; meniscus contact height and dip; tube top;
  // pool surface and floor.
  let (r, ry, m, dip, top, level, floor) = (1, .4, 3, 1, 4, .25, -1.8)
```

```

rect((-3 * r, floor), (3 * r, level), fill: water, stroke: none)
line((-3 * r, level + .1), (-3 * r, floor), (3 * r, floor), (3 * r, level + .1))
rect((-r, 0), (r, m), fill: water, stroke: none)

for (side, x) in (("left", -r), ("right", r)) {
  line((x, 0), (x, top), name: side)
  line((x, 0), (rel: (0, -.3)), stroke: (dash: "dashed"))
  line((x, top), (rel: (0, .3)), stroke: (dash: "dashed"))
}
circle((0, top), radius: (r, ry), stroke: (dash: "dashed"))

// A quadratic Bézier is an exact parabola; its control point lies on both contact tangents.
anchor("ctrl", (0, m - 2 * dip))
anchor("surface", (0, m))
anchor("contact", (r, m))
let rim = arc.with("surface", radius: (r, ry), anchor: "origin", start: 0deg)
circle("surface", radius: (r, ry), fill: blue.lighten(55%), stroke: none)
rim(delta: 180deg, stroke: (dash: "dashed"))
merge-path(fill: blue.lighten(70%), {
  rim(delta: -180deg)
  bezier((-r, m), "contact", "ctrl")
})

// Surface tension acts along the contact tangent, all around the rim.
for a in range(0, 360, step: 45) {
  let a = a * 1deg
  arrow((rel: (a, (r, ry)), to: "surface"), (rel: (.5 * calc.cos(a), 1 + .2 * calc.sin(a))))
}
line("contact", ("contact", 50%, "ctrl"))
cetz.angle.angle("contact", "right.start", "ctrl", direction: "near", radius: .9,
  label: $alpha$, label-radius: 125%, mark: (symbol: "stealth", scale: .5),
  name: "alpha")
callout((rel: (.15, 0), to: "alpha.label"), (rel: (.9, .6), to: "alpha.label"),
  [angle between the \ meniscus and the \ tube wall])

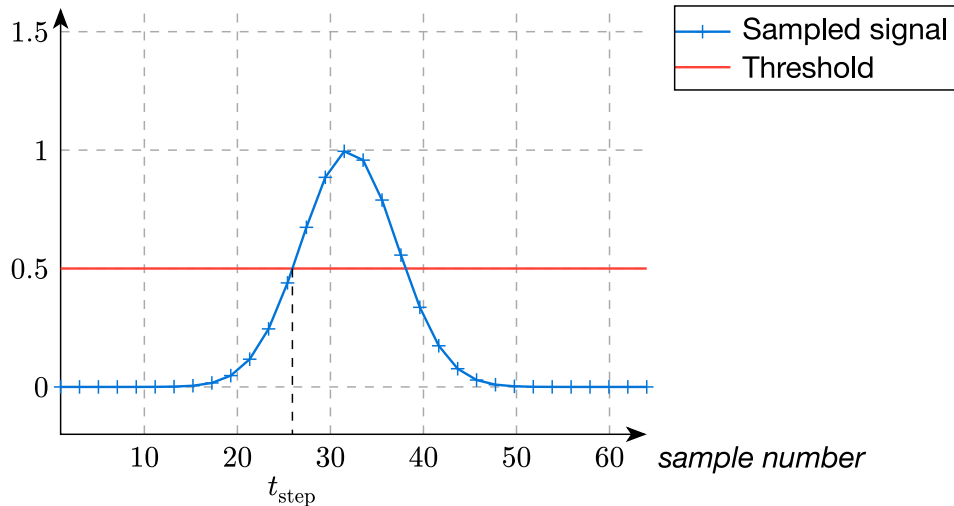
arrow((0, m - dip - .4), (rel: (0, -1)), stroke: 3pt, name: "weight")
content("weight.end", $pi r^2 h rho g$, anchor: "north")
arrow((0, top + .8), (rel: (0, 1)), stroke: 3pt, name: "lift")
content("lift.end", $2 pi r gamma cos(alpha)$, anchor: "south")

brace((r + .2, m - dip), (r + .2, level), name: "h")
content("h.content", $h$, anchor: "west")
callout((rel: (0, -.25), to: "h.content"), (rel: (.6, -.6), to: "h.content"), [rise],
  out: -90deg)
brace((0, -.4), (-r, -.4), name: "r")
content("r.content", $r$, anchor: "north")
callout((rel: (.2, -.2), to: "r.content"), (rel: (.8, -.5), to: "r.content"), [tube radius])
}

```

Threshold crossing

Redrawn from [anotacoes_intersecao.pg](#) (png) in Walmes M. Zeviani's *Tikz Gallery*, which is based on [an answer by Red](#) on TeX Stack Exchange (CC BY-SA 3.0). `intersections` finds where the signal crosses the threshold, and a `"|-"` coordinate drops it to a computed tick; standalone marks on data points; a legend boxed with `rect-around`.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

// Based on https://tex.stackexchange.com/a/130791 by Red, CC BY-SA 3.0.

#set canvas({
  import cetz.draw: *
  set-style(stroke: .5pt + black, content: (padding: 4pt))
  // Data coordinates (sample, amplitude); text, strokes and marks keep their size.
  scale(x: .11, y: 2.8)

  let (x0, x1, y0, y1) = (1, 64, -.2, 1.6)
  let signal(x) = calc.exp(-calc.log(2) / 16 * calc.pow(x - 32, 2))
  let samples = range(32).map(i => x0 + (x1 - x0) * i / 31).map(x => (x, signal(x)))

  for x in range(10, 61, step: 10) {
    line((x, y0), (x, y1), stroke: (paint: gray, dash: "dashed"))
    content((x, y0), $x$, anchor: "north")
  }
  for y in (0, .5, 1, 1.5) {
    line((x0, y), (x1, y), stroke: (paint: gray, dash: "dashed"))
    content((x0, y), $y$, anchor: "east")
  }
  line((x0, y1), (x0, y0), (x1, y0), mark: (start: "stealth", end: "stealth", fill: black))
  content((x1, y0), emph[sample number], anchor: "north-west")

  line((x0, .5), (x1, .5), stroke: .8pt + red, name: "threshold")
  let plus(p) = mark(p, 0deg, "+", stroke: blue, length: .16, width: .16)
  line(..samples, stroke: .8pt + blue, name: "signal")
  samples.map(plus).join()

  // "cross.0" is the rising crossing, the first one along the signal.
  intersections("cross", "signal", "threshold")
  line("cross.0", ("cross.0", "|-", (0, y0)), stroke: (dash: "dashed"), name: "drop")
  content((rel: (0, -.12), to: "drop.end"), $t_"step"$, anchor: "north")
})
```

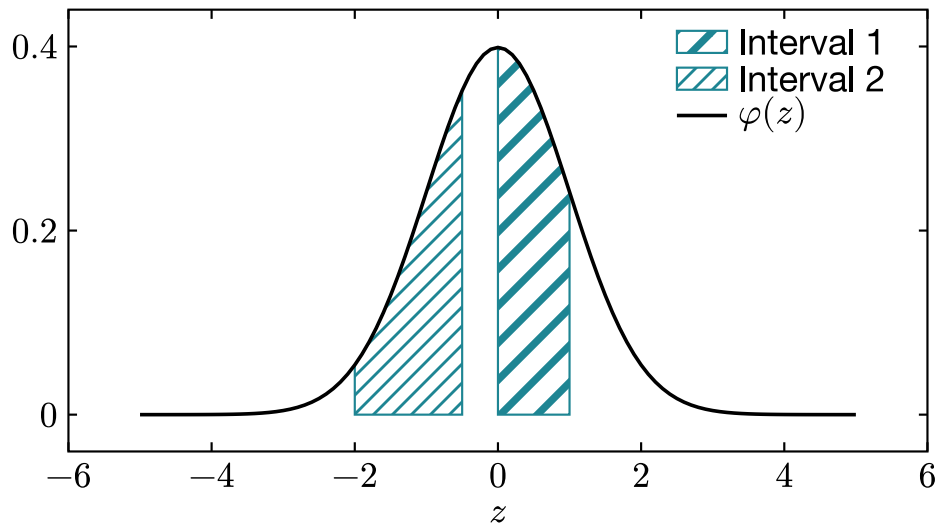
```

group(name: "legend", {
  set-origin((x1 + 3, y1))
  for (i, (hue, label)) in ((blue, [Sampled signal]), (red, [Threshold])).enumerate() {
    line((0, -.15 * i - .1), (rel: (6, 0)), stroke: .8pt + hue, name: "key")
    if i == 0 { plus("key.50%") }
    content("key.end", label, anchor: "west")
  }
})
rect-around("legend")
})

```

Hatched intervals

Redrawn from [preenchimento.pgf \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*, which is based on [an answer by Jake](#) on TeX Stack Exchange (CC BY-SA 3.0). Typst tiling hatches with spacing and thickness as parameters; data units via scale; one loop for each series and its legend row.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)
// Based on https://tex.stackexchange.com/a/29367 by Jake, CC BY-SA 3.0.

#let hue = eastern.darken(15%)
// Diagonal hatch with period `d`; lines overshoot the tile so that butt caps do not notch the
// corners.
#let hatch(d, thickness) = tiling(size: (d, d), for c in (0pt, d, 2 * d) {
  place(line(start: (c + d, -d), end: (-d, c + d), stroke: thickness + hue))
})

#cetx.canvas({
  import cetx.draw: *
  set-style(stroke: .6pt + black)
  // Data coordinates (z, φ(z)).
  scale(x: .7, y: 9)

  let pdf(z) = calc.exp(-z * z / 2) / calc.sqrt(2 * calc.pi)
  let curve(a, b) = range(101).map(i => a + (b - a) * i / 100).map(z => (z, pdf(z)))

  rect((-6, -.04), (6, .44), name: "frame")
  for x in range(-6, 7, step: 2) {
    let p = ((x, 0), "|-", "frame.south")
    line(p, (rel: (0, .01)))
    line((p, "|-", "frame.north"), (rel: (0, -.01)))
    content(p, $#x$, anchor: "north", padding: 3pt)
  }
  for y in (0, .2, .4) {
    let p = ("frame.west", "|-", (0, y))
    line(p, (rel: (.12, 0)))
    line((p, "-|", "frame.east"), (rel: (-.12, 0)))
    content(p, $#y$, anchor: "east", padding: 3pt)
  }
  content((rel: (0, -.05), to: "frame.south"), $z$, anchor: "north")

  // Each series with its legend row.
```

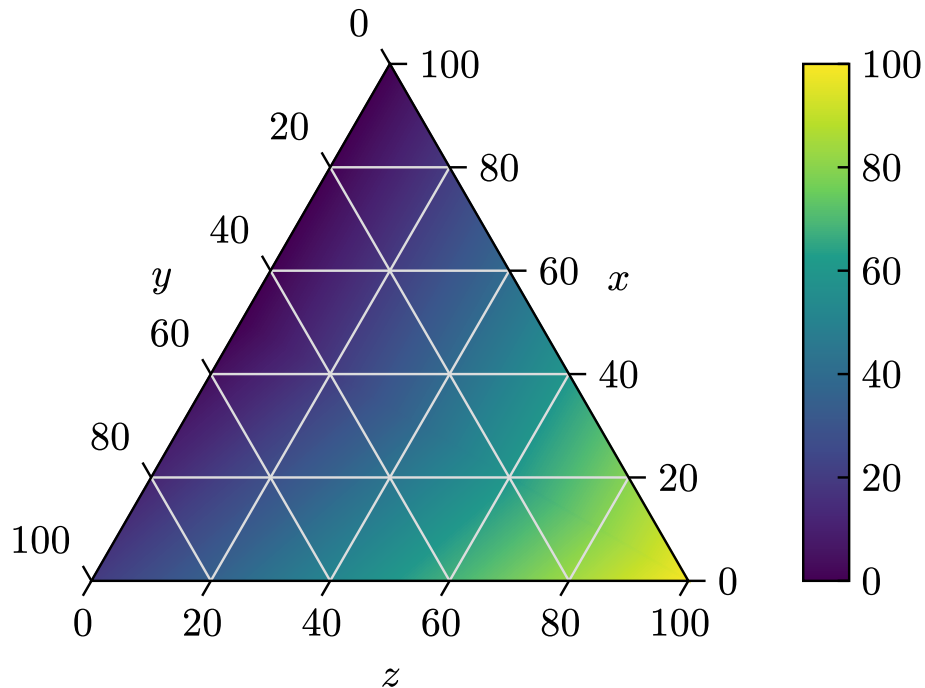
```

for (i, (fill, label, (a, b))) in (
  (hatch(10pt, 2pt), [Interval 1], (0, 1)),
  (hatch(5pt, .75pt), [Interval 2], (-2, -.5)),
  (none,  $\phi(z)$ , (-5, 5)),
).enumerate() {
  let p = (rel: (-2.8, -.035 - .04 * i), to: "frame.north-east")
  if fill == none {
    line(..curve(a, b), stroke: 1pt)
    line((rel: (-.7, 0), to: p), p, stroke: 1pt)
  } else {
    line((a, 0), ..curve(a, b), (b, 0), close: true, fill: fill, stroke: hue)
    rect((rel: (-.7, -.012), to: p), (rel: (0, .012), to: p), fill: fill, stroke: hue)
  }
  content(p, label, anchor: "west", padding: 3pt)
}
})

```

Ternary diagram

Redrawn from [ternario.pgf \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*, which is based on [an answer by Christian Feuersänger](#) on TeX Stack Exchange (CC BY-SA 3.0). Barycentric coordinates; each triangle filled with an exact gradient.linear sampled from `color.map.viridis`; one loop for the grid, ticks and labels of all three sides; a matching color bar.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)
// Based on https://tex.stackexchange.com/a/277000 by Christian Feuersänger, CC BY-SA 3.0.

#let cmap = gradient.linear(..color.map.viridis, angle: -90deg)
#let keys = ("x", "y", "z")
#let corners = range(3).map(i => 90deg + i * 120deg).map(a => (3 * calc.cos(a), 3 *
calc.sin(a)))

#cetz.canvas({
  import cetz.draw: *
  import cetz: vector
  set-style(stroke: .6pt + black, content: (padding: 3pt))
  for (k, c) in keys.zip(corners) { anchor(k, c) }

  // Gouraud-shade a triangle given as rows (x, y, z, value) of the source table.
  let patch(..rows) = {
    let rows = rows.pos()
    let pts = rows.map(r => corners.zip(r).map(((c, w)) => vector.scale(c,
w)).reduce(vector.add))
    let vs = rows.map(r => r.last())
    // Gradient of the linear interpolant, from the sides rotated a quarter turn.
    let normals = range(3).map(i => vector.sub(pts.at(calc.rem(i + 1, 3)), pts.at(calc.rem(i +
2, 3))))
    .map(((dx, dy)) => (dy, -dx))
    let det = range(3).map(i => pts.at(i).at(0) * normals.at(i).at(0)).sum()
    let g = range(3).map(i => vector.scale(normals.at(i), vs.at(i) / det)).reduce(vector.add)
    // Typst turns gradients clockwise and stretches them between the extreme bounding-box
```

```

corners.
  let (xs, ys) = array.zip(..pts)
  let vc = for x in (calc.min(..xs), calc.max(..xs)) { for y in (calc.min(..ys),
calc.max(..ys)) {
    (vs.first() + vector.dot(g, vector.sub((x, y), pts.first())),)
  } }
  let (lo, hi, va, vb) = (calc.min(..vs), calc.max(..vs), calc.min(..vc), calc.max(..vc))
  let stops = range(11).map(k => lo + (hi - lo) * k / 10)
    .map(v => (cmap.sample(v * 1%), (v - va) / (vb - va) * 100%))
  let paint = gradient.linear((stops.first().at(0), 0%), ..stops, (stops.last().at(0), 100%),
    angle: -calc.atan2(..g))
  line(..pts, close: true, fill: paint, stroke: .5pt + paint)
}
patch((0, 0, 1, 100), (1, 0, 0, 0), (.5, .5, 0, 0))
patch(.5, .5, 0, 0), (0, 1, 0, 20), (0, 0, 1, 100))

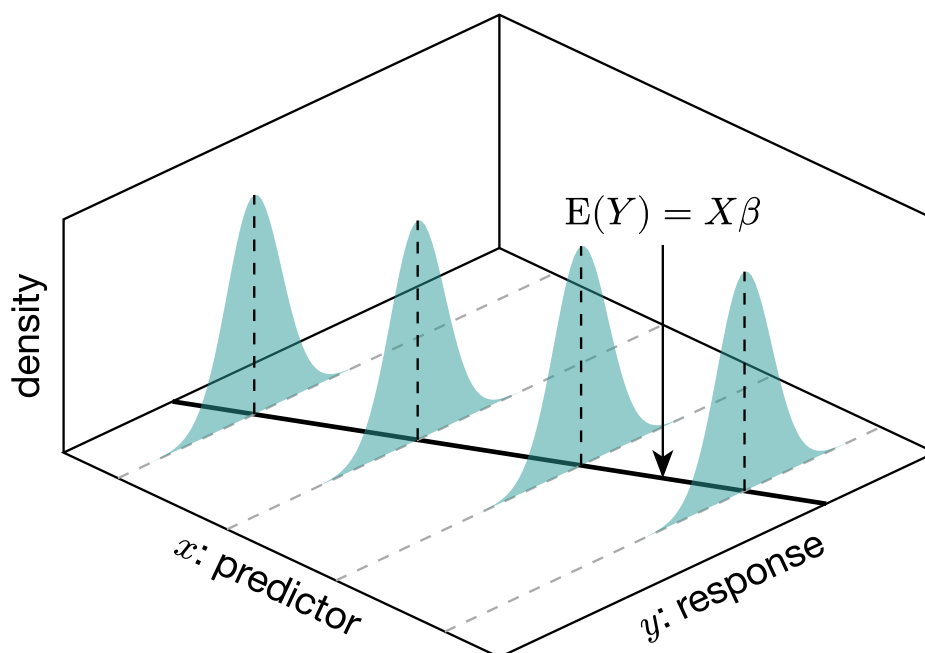
// Axis k is read along the side where the next key is zero; its lines of constant k
// run from that side towards the side where the key after next is zero.
for (i, k) in keys.enumerate() {
  let (a, b) = (keys.at(calc.rem(i + 1, 3)), keys.at(calc.rem(i + 2, 3)))
  let dir = i * 120deg
  for v in range(0, 101, step: 20) {
    let t = v / 100
    let edge = (bary: ((k): t, (b): 1 - t))
    if 0 < v and v < 100 { line((bary: ((k): t, (a): 1 - t)), edge, stroke: silver) }
    line(edge, (rel: (dir, .15)), name: "tick")
    content("tick.end", $#v$, anchor: ("west", "south-east", "north").at(i))
  }
  content((rel: (dir + 30deg, .8), to: (bary: ((k): 1, (b): 1))), $italic(#k)$)
}
line(..keys, close: true)

rect((rel: (1, 0), to: "z"), ((rel: (1.4, 0), to: "z"), "|-", "x"), fill: cmap,
  name: "bar")
for v in range(0, 101, step: 20) {
  let p = ("bar.south-east", v * 1%, "bar.north-east")
  line(p, (rel: (-.1, 0)))
  content(p, $#v$, anchor: "west")
}
})

```

Regression densities in 3D

Redrawn from [reg_model_3d.png](#) in Walmes M. Zeviani's *Tikz Gallery*, which is based on [an answer by Jake](#) on TeX Stack Exchange (CC BY-SA 3.0). An ortho view drawn back to front; on-xz for the floor and on-xy for density curves standing on it; labels rotated to the projected edges.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)
// Based on https://tex.stackexchange.com/a/53795 by Jake, CC BY-SA 3.0.

#let normal(y, mu) = calc.exp(-calc.pow(y - mu, 2) / 2) / (2 * calc.sqrt(calc.pi))

#cetz.canvas({
  import cetz.draw: *
  let elev = 28deg
  // Screen slope of the floor edges, used to lay labels along them.
  let tilt = calc.atan(calc.sin(elev))
  let (lo, hi, top) = (-4, 12, 0.3)

  // Canvas x = response y, canvas z = predictor x, canvas y = density.
  ortho(x: elev, y: -45deg, sorted: false, {
    scale(x: 0.35, y: 8, z: 0.7)
    set-style(stroke: 0.6pt + black)

    on-xz(rect((lo, 0), (hi, 8)))
    line((lo, 0, 0), (lo, top, 0), (hi, top, 0), (hi, top, 8), (hi, 0, 8))
    line((hi, 0, 0), (hi, top, 0))
    on-xz(line((0, 0), (8, 8), stroke: 1.2pt))

    for x in (1, 3, 5, 7) {
      on-xz(line((lo, x), (hi, x), stroke: (paint: gray, dash: "dashed")))
      on-xy(z: x, {
        let ys = range(61).map(i => x - 4 + i * 8 / 60)
        line(ys.map(y => (y, normal(y, x))), close: true, stroke: none,
          fill: teal.darken(25%).transparentize(50%))
        line((x, 0), (x, normal(x, x)), stroke: (dash: "dashed"))
      })
    }
  })
}
```

```

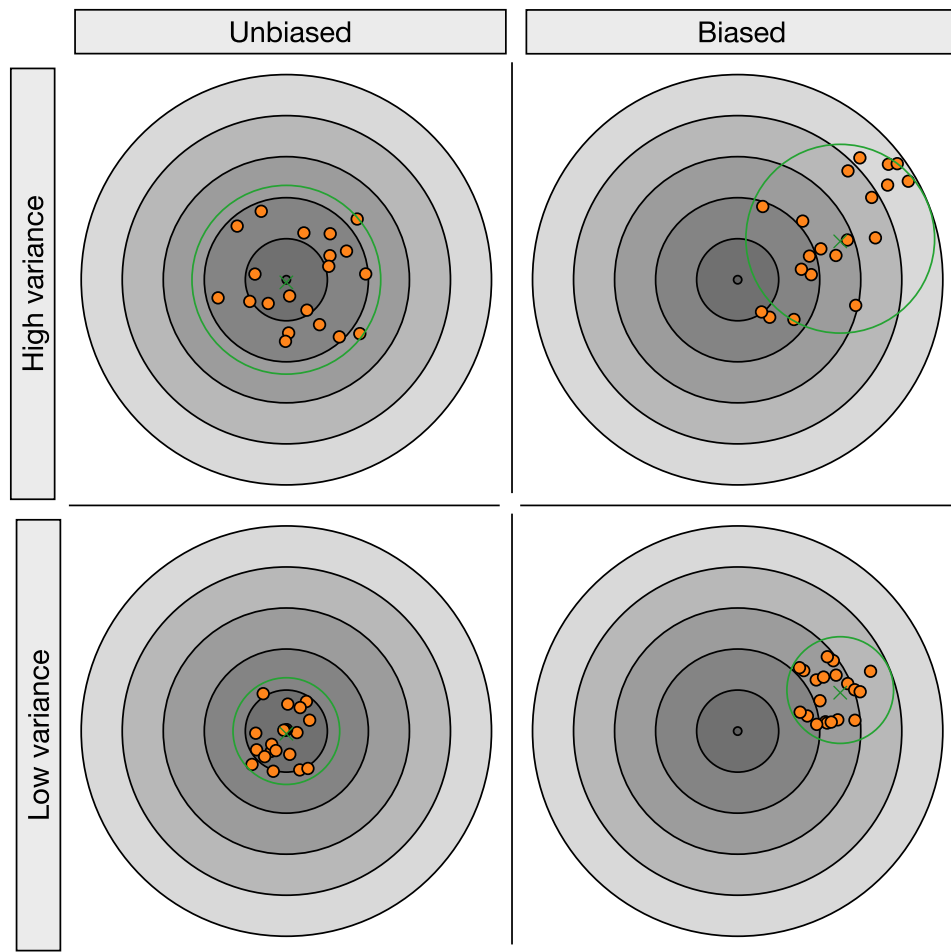
on-xy(z: 6, {
  line((6, top), (6, 0), mark: (end: "stealth", fill: black))
  content((6, top), $upright(E)(Y) = X beta$, anchor: "south", padding: 0.1)
})

content((lo, 0, 4), [$x$: predictor], angle: -tilt, anchor: "north", padding: 0.15)
content((4, 0, 8), [$y$: response], angle: tilt, anchor: "north", padding: 0.15)
content((lo, top / 2, 0), [density], angle: 90deg, anchor: "south", padding: 0.15)
})
})

```

Bias and variance targets

Redrawn from [bias-variance.pgf \(png\)](#) in Walmes M. Zeviani's *Tikz Gallery*. Small multiples from one parameterized panel in nested loops; seeded pseudo-random hits from a tiny inline generator; rotated row headers.



```
#import "@preview/cetz:0.5.2"
#set page(width: auto, height: auto, margin: 4mm)
#set text(font: ("Helvetica Neue", "Arial"), size: 10pt, fill: black)

// Linear congruential generator mapped to [-1, 1); Typst has no random numbers.
#let uniform(seed, n) = {
  let m = calc.pow(2, 31)
  let s = seed
  for _ in range(n) {
    s = calc.rem(1103515245 * s + 12345, m)
    (2 * s / m - 1,)
  }
}

#cetz.canvas({
  import cetz.draw: *
  set-style(stroke: 0.6pt + black)

  let panel(name, origin, aim, spread, ring, seed) = group(name: name, {
    translate(origin)
    for r in (2.5, 2, 1.5, 1, 0.5, 0.05) {
      circle((0, 0), radius: r, fill: black.transparentize(85%))
    }
  })
}
```

```

    for (dx, dy) in uniform(seed, 40).chunks(2) {
        circle((rel: (spread * dx, spread * dy), to: aim), radius: 2pt, fill: orange)
    }
    circle(aim, radius: ring, stroke: green.darken(20%))
    content(aim, text(green.darken(30%), $times$))
    line((-2.65, -2.75), (rel: (5.25, 0)))
    line((2.75, 2.65), (rel: (0, -5.25)))
})
let header(pos, body, ..args) = content(pos, ..args, padding: 0.1,
    box(width: 5.25cm, inset: 4pt, fill: silver.lighten(40%), stroke: 0.6pt, align(center,
body)))

let cols = (([Unbiased], (0, 0)), ([Biased], (1.25, 0.5)))
let rows = (([High variance], 1, 1.15), ([Low variance], 0.5, 0.65))
for (j, (row, spread, ring)) in rows.enumerate() {
    for (i, (col, aim)) in cols.enumerate() {
        let name = "p" + str(i) + str(j)
        panel(name, (5.5 * i, -5.5 * j), aim, spread, ring, 1 + 2 * j + i)
        if i == 0 { header(name + ".west", row, angle: 90deg, anchor: "south") }
        if j == 0 { header(name + ".north", col, anchor: "south") }
    }
}
})

```